Problem Set I

Econometrics I - FGV EPGE

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Problem 1 – (points: 1)

Let X be a scalar random variable with density $f(x)$. Let $K(\cdot)$ be a symmetric second-order kernel. For a given point x in the interior of the support of $f(\cdot)$, define the density estimator as

$$\hat{f}_n(x) \equiv \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right),$$

where $h > 0$ is a bandwidth parameter and $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables with density $f(\cdot)$.

This exercise will show you how to ensure that $\hat{f}_n(x) \xrightarrow{p} f(x)$ as $n \to \infty$, $h \to 0$, and $nh \to \infty$. This is the same asymptotic framework as in the slides.

a) Show that $\hat{f}_n(x) \geq 0$ for all x, and that $\int_{-\infty}^{\infty} \hat{f}_n(x)dx = 1$ for all n.

b) Assume from now on that the density f is bounded on $\mathbb{R}$, continuous at x, and $f(x) > 0$. Show that $\mathbb{E}[\hat{f}_n(x)] = f(x) + o(1)$.

c) Show that $\text{Var}(\hat{f}_n(x)) = \frac{1}{nh} \cdot f(x)R(K) + o\left(\frac{1}{nh}\right)$, where $R(K) = \int_{-\infty}^{\infty} K^2(u)du$.

d) Argue that these results imply that $\hat{f}_n(x)$ is consistent for $f(x)$.

e) Now, assume that f is twice differentiable in a neighborhood of x, with f'' continuous at x, and that K has unit variance, $\int_{-\infty}^{\infty} u^2 K(u)\, du = 1$. Show that

$$\mathbb{E}[\hat{f}_n(x)] = f(x) + \frac{h^2}{2} f''(x) + o(h^2)$$

f) Explain in words how the local convexity of f might (or might not) affect this finite-sample bias.

Hint: A very useful resource for this question is Chapter 17 from *Probability and Statistics for Economists* by Bruce Hansen.

Problem 2 – (points: 1)

Consider the nonparametric regression model:

$$Y_i = m(X_i) + u_i, \quad i = 1, \ldots, n$$

where $\mathbb{E}[u_i|X_i] = 0$ and $m(\cdot)$ is an unknown smooth function. The **local-linear estimator** of $(m(x_0); m'(x_0))$ at a point x_0 is defined as the solution to the weighted least squares problem:

$$\{m_{LL}(x_0), m'_{LL}(x_0)\} \equiv \arg\min_{\alpha,\beta} \sum_{i=1}^{n} K_h(X_i - x_0) \left[Y_i - \alpha - \beta(X_i - x_0)\right]^2$$

where $K_h(u) = \frac{1}{h}K\left(\frac{u}{h}\right)$ is a second-order continuous kernel function with bandwidth $h > 0$. You have a random sample of size n: $\{(Y_1, X_1), ..., (Y_n, X_n)\}$. We also assume that $X_i \in \mathbb{R}$.

a) Show that the local-linear estimator can be written as a Generalized Least Squares (GLS) estimator. Specifically, show that there exists an $n \times 1$ vector V, an $n \times 2$ matrix Z, and an $n \times n$ matrix W such that:

$$\begin{pmatrix} m_{LL}(x_0) \\ m'_{LL}(x_0) \end{pmatrix} = (Z'W^{-1}Z)^{-1}Z'W^{-1}V$$

b) What happens to this estimator if, for a given n, we have that $h \to \infty$?

c) When deriving the asymptotic theory for the local-constant estimator, we assumed the bandwidth was such that $h \to 0$ and $nh \to \infty$ as $n \to \infty$. Explain intuitively why each of these two conditions is necessary.

d) Are such conditions also needed for the asymptotic theory of the local-linear estimator? What are the main differences between these two estimators? Briefly explain.

Problem 3 – (points: 3)

This is an empirical question based on Karlan and Zinman (2008), "Credit Elasticities in Less-Developed Economies: Implications for Microfinance", *American Economic Review*. The data, `karlan.csv`, is in the `problem_sets/PS1` folder of the class GitHub repo.

a) What is the main research question in the paper? What is the most striking finding? Answer in just a few sentences.

b) Your goal will be to estimate $\mathbb{P}(applied = 1|offer4 = x)$. Note that `applied` is a binary variable, while `offer4` is continuous. These are the only variables you will need. Notice that

$$\mathbb{P}(applied = 1|offer4 = x) = \mathbb{E}[applied|offer4 = x]$$

c) Use a Gaussian kernel to estimate this probability and show a plot of your estimates for a range of values of `offer4`. Do this with three different bandwidths:

- Silverman's rule of thumb: $h = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5}$, where $\hat{\sigma}$ is the standard deviation of `offer4` and n is the number of observations;
- A value much *smaller* than that;
- A value much *larger* than that;

d) Do the same with the Epanechnikov kernel, using the same bandwidths as before.

e) Compare the results of the two kernels qualitatively.

Problem 4 – (points: 1)

Let $X_i, i = 1, \ldots, n$ be an i.i.d. sample of observations with distribution $U[0, \theta]$. A natural estimator (in fact, the MLE) of θ is $X_{(n)}$, where

$$\min\{X_1, ..., X_n\} = X_{(1)} \leq \cdots \leq X_{(n)} = \max\{X_1, ..., X_n\}$$

denote the ordered values of the data. These statistics are sometimes referred to as the order statistics of the data. Consider the following root:

$$R_n = n(X_{(n)} - \theta),$$

a) Show that $R_n \leq 0$.

b) Let $J_n(x, P) = P(R_n \leq x)$ be the cdf of the root R_n above. Show that $J_n(x, P)$ converges in distribution to $J(x, P) = P(-\theta X \leq x)$, where $X \sim \exp(1)$, i.e.,

$$J(x, P) = \begin{cases} e^{x/\theta}, & x \leq 0 \\ 1 & \text{otherwise} \end{cases}$$

Hint: for every $r \in \mathbb{R}$, remember that $\left(1 + \frac{r}{n}\right)^n \xrightarrow{n \to \infty} e^r$.

c) We will now prove that the bootstrap will not work in this case. Consider a non-random sequence of probability distributions $\{P_n\}_{n=1}^{\infty}$ such that each distribution P_n puts equal mass on n distinct points in the interval $[0, \theta]$.

For each n, let $\{X_{i,n}\}_{i=1}^{n}$ be an i.i.d. sample from P_n. Denote by $X_{(n),n}$ the maximum of this n-th sample. Also, denote by $\theta(P_n)$ the maximum of the support of P_n.

Now, show that $\lim_{n \to \infty} P\left(n(X_{(n),n} - \theta(P_n)) \leq -\epsilon\right) \leq e^{-1}, \forall \epsilon > 0$.

d) Use this result to show that $J_n(x, P_n)$, which is the distribution of the bootstrapped root under the measure P_n does *not* converge to $J(x, P)$.

Hint: pick $\epsilon < \theta$.

e) Argue, using the last item, that the bootstrap does not work in this case. Think about the relationship between the empirical distribution and the sequence of distributions $\{P_n\}_{n=1}^{\infty}$.

Problem 5 – (points: 2)

This question is about the bootstrap. Suppose that $y_i = \alpha + \beta x_i + \epsilon_i$, where $\alpha = 2$, $\beta = 5$, $\epsilon_i | x_i \sim N(0, 2 + x_i^2)$, and $x \sim N(0, 3)$.

a) Compute the conditional distribution of y_i given x_i;

b) Simulate a random sample of size $n = 100$ and create a scatter plot of your data;

c) Estimate the parameters using OLS;

d) Report the estimates and three different symmetric confidence intervals for β using different measures of standard errors:

- The usual OLS standard error that ignores heteroskedasticity;
- A heteroskedasticity-robust standard error;
- A bootstrap standard error, using $B = 1000$ bootstrap samples You should also plot the histogram of $\hat{\beta}$ computed from the bootstrap samples.

It will be helpful for the next item if you structure your code so that it receives a certain sample $\{(y_1, x_1), (y_2, x_2), ..., (y_n, x_n)\}$ and spits out the three different confidence intervals.

e) Repeat the steps from items b), c), and d) for $m = 500$ times and check how frequently the three different confidence intervals cover the true value of $\beta = 5$. You can use the same code from item b) to do this, but you will need to modify it so that it runs m times and stores the results.

Problem 6 – (points: 2)

This question connects the bootstrap with the kernel density estimator from Problem 1. It is mostly about coding and explaining what you find. Let $X_1, \ldots, X_n$ be i.i.d. $N(0, 1)$, so the true density is $f = \phi$, the standard normal density. At a point x, estimate $f(x)$ with the estimator from Problem 1,

$$\hat{f}_n(x) \equiv \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right),$$

using the Gaussian kernel, $K = \phi$, and Silverman's rule of thumb, $h = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5}$, where $\hat{\sigma}$ is the sample standard deviation.

a) Write a function that receives a sample $\{X_1, \ldots, X_n\}$ and a point x, and returns a 95% bootstrap confidence interval for $f(x)$ based on the root $R_n = \hat{f}_n(x) - f(x)$. Specifically:

   - Draw B bootstrap samples (with replacement) from the data, where B is the number of bootstrap replications;
   - In each bootstrap sample $b = 1, \ldots, B$, recompute the estimate $\hat{f}^*_{n,b}(x)$, using the *same* bandwidth h as in the original sample, and store the bootstrap root $R^*_{n,b} = \hat{f}^*_{n,b}(x) - \hat{f}_n(x)$;
   - Let L_n be the empirical CDF of the B bootstrap roots, and report the interval

$$\left[\hat{f}_n(x) - L_n^{-1}(0.975),\ \hat{f}_n(x) - L_n^{-1}(0.025)\right].$$

b) Set $n = 500$. Simulate $m = 1{,}000$ samples and, for each one, compute the confidence interval from item a) at $x = 0$ and at $x = 1$. Report how often each interval covers the true value $f(x)$.

c) Explain the difference in coverage between $x = 0$ and $x = 1$.

   Hint: compute $f''(x)$ for the standard normal density and recall item e) from Problem 1.

d) Let $\mathbb{E}^*[\cdot]$ denote the expectation over bootstrap samples, holding the original data fixed, and let $\hat{f}^*_n(x)$ be the estimate computed from one bootstrap sample. Show that $\mathbb{E}^*[\hat{f}^*_n(x)] = \hat{f}_n(x)$. What does this tell you about the ability of the bootstrap to detect the bias of $\hat{f}_n(x)$?

e) Redo item b) at $x = 0$ with $n = 2{,}000$. Does more data fix the coverage? Explain why, comparing how the bias and the standard deviation of $\hat{f}_n(x)$ shrink with n when h follows Silverman's rule.

f) Go back to $n = 500$ and $x = 0$, but now use a smaller bandwidth, $h/3$, both in the original sample and in the bootstrap samples. What happens to the coverage and to the average width of the confidence interval? Explain the trade-off.